

A Short Mathematical Reference for the q-State TSP-QEFEM Formulation

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Abstract

This note records the mathematical formulation used for the corrected q-state Quantum-Enhanced Free-Energy Machine (QEFEM) implementation for the Traveling Salesman Problem (TSP), with Berlin52 as the reference instance. The purpose is to state the variables, functions, free-energy objective, annealing schedule, optimization sequence, and decoding rule in a concise proof-style form. The formulation treats each tour slot as a q-state quantum system, where q equals the number of cities. Therefore the local object is a q-dimensional density matrix, not a two-level Bloch vector.

Contents

1 Problem Statement	3
2 Slot Variables and Mean-Field Interpretation	3
3 Classical TSP Mean-Field Terms	4
3.1 Expected Tour Length	4
3.2 City-Usage Penalty	4
4 q-State Quantum Construction	4
4.1 Trainable Diagonal Energies	4
4.2 Distance-Aware Mixer	5
4.3 Local Quantum Hamiltonian	5
4.4 Local Gibbs Density Matrix	5
4.5 Eigendecomposition Form	5
5 Quantum Entropy and Mixer Energy	6
5.1 Quantum Entropy	6
5.2 Mixer Expectation and Mixer Energy	6
6 Corrected TSP-QEFEM Free Energy	6
7 Hyperparameters Used in the Corrected Berlin52 Run	7
8 Annealing Schedules	7
9 Chronological Algorithm	8
10 Decoding Rule	8
11 Result of the Corrected Run	9

12 Key Takeaways**9**

1 Problem Statement

Let there be N cities. For Berlin52, $N = 52$. Let

$$D_{ij}^{\text{raw}} \geq 0$$

denote the TSPLIB integer Euclidean distance between city i and city j , and let

$$D_{ij} = \frac{D_{ij}^{\text{raw}}}{\max_{a,b} D_{ab}^{\text{raw}}}$$

denote the normalized distance used during optimization. The raw matrix is retained for final evaluation.

A TSP tour is a permutation

$$\pi = (\pi_1, \pi_2, \dots, \pi_N)$$

of the city labels $\{1, 2, \dots, N\}$. The raw tour length is

$$L_{\text{raw}}(\pi) = \sum_{t=1}^N D_{\pi_t, \pi_{t+1}}^{\text{raw}}, \quad \pi_{N+1} = \pi_1. \quad (1)$$

The normalized tour length is obtained by replacing D^{raw} with D .

For Berlin52, the known optimum is

$$L_{\text{opt}} = 7542.$$

The gap percentage of a found tour π is computed as

$$\text{gap}(\pi) = 100 \cdot \frac{L_{\text{raw}}(\pi) - 7542}{7542}. \quad (2)$$

2 Slot Variables and Mean-Field Interpretation

The TSP is represented using N tour slots. Slot t chooses one city from $\{1, \dots, N\}$. Thus each slot is a q -state variable with

$$q = N.$$

For a classical mean-field formulation, one would use probabilities

$$P_{t,c} = \Pr(\text{city } c \text{ is assigned to slot } t), \quad \sum_{c=1}^N P_{t,c} = 1.$$

For the replica formulation, the probability becomes

$$P_{r,t,c},$$

where $r = 1, \dots, R$ indexes replicas.

Remark 1 (Why there is no R, θ Bloch parametrization). *The variables R and θ are natural for a two-state qubit or binary MaxCut-type formulation, where a local density matrix can be written in terms of Bloch components. In TSP, each slot has $N = 52$ possible city states. Therefore the local quantum object is a 52×52 density matrix. A two-dimensional Bloch parametrization is not the correct representation for this q -state formulation.*

3 Classical TSP Mean-Field Terms

3.1 Expected Tour Length

Given a replica probability tensor $P_{r,t,c}$, define the next-slot probability by cyclic shift:

$$P_{r,t,d}^{\text{next}} = P_{r,t+1,d}, \quad P_{r,N+1,d} = P_{r,1,d}.$$

The expected normalized tour length is

$$U_{\text{tour}}^{(r)}(P) = \sum_{t=1}^N \sum_{c=1}^N \sum_{d=1}^N P_{r,t,c} D_{cd} P_{r,t+1,d}. \quad (3)$$

Proposition 1. *If the slot assignments are independent under the mean-field distribution, then $U_{\text{tour}}^{(r)}(P)$ is the expected normalized cyclic tour length.*

Proof. For a fixed edge from slot t to slot $t+1$, the probability that city c is assigned to slot t and city d is assigned to slot $t+1$ is

$$P_{r,t,c} P_{r,t+1,d}$$

by the mean-field independence assumption. The expected distance contribution of this edge is therefore

$$\sum_{c,d} P_{r,t,c} D_{cd} P_{r,t+1,d}.$$

Summing this over all cyclic slots $t = 1, \dots, N$ gives the stated formula. $\square$

3.2 City-Usage Penalty

A valid TSP tour must use every city exactly once. In the soft relaxation, define the city usage

$$C_{r,c}(P) = \sum_{t=1}^N P_{r,t,c}. \quad (4)$$

The penalty term is

$$U_{\text{city}}^{(r)}(P) = \lambda_{\text{city}} \sum_{c=1}^N (C_{r,c}(P) - 1)^2. \quad (5)$$

Proposition 2. *For an integral assignment matrix $P_{r,t,c} \in \{0, 1\}$ with exactly one city per slot, $U_{\text{city}}^{(r)} = 0$ if and only if each city is used exactly once.*

Proof. If each city is used exactly once, then $C_{r,c} = 1$ for all c , so every squared term vanishes. Conversely, if the penalty is zero, every squared term must vanish, hence $C_{r,c} = 1$ for all cities c . Since every slot already contains exactly one city, this condition means no city is missing and no city is repeated. $\square$

4 q-State Quantum Construction

4.1 Trainable Diagonal Energies

For each replica r , slot t , and city c , introduce a trainable scalar

$$a_{r,t,c} \in \mathbb{R}.$$

The vector $a_{r,t} = (a_{r,t,1}, \dots, a_{r,t,N})$ defines the diagonal part of a local Hamiltonian.

4.2 Distance-Aware Mixer

The corrected implementation uses a distance-aware symmetric mixer. First define

$$K_{cd} = \begin{cases} \exp(-D_{cd}/\tau), & c \neq d, \\ 0, & c = d, \end{cases} \quad (6)$$

where $\tau > 0$ is the mixer length scale. Then define the degree

$$d_c = \sum_{d=1}^N K_{cd},$$

and construct the symmetric normalized matrix

$$\widetilde{M}_{cd} = d_c^{-1/2} K_{cd} d_d^{-1/2}. \quad (7)$$

Finally, normalize by its spectral radius:

$$M = \frac{\widetilde{M}}{\rho(\widetilde{M})}, \quad \rho(\widetilde{M}) = \max_k |\lambda_k(\widetilde{M})|. \quad (8)$$

This keeps M symmetric and gives it a controlled numerical scale.

Remark 2 (Reason for changing the mixer). *A complete-graph uniform mixer has a highly degenerate eigenspectrum. Since the training requires gradients through an eigendecomposition, near-degenerate eigenvalues can lead to numerical instability. The distance-aware mixer breaks the large degeneracy while preserving symmetry and giving the mixer a TSP-relevant structure.*

4.3 Local Quantum Hamiltonian

For every replica and slot, define the local q-state Hamiltonian

$$H_{r,t} = \text{diag}(a_{r,t}) - \gamma M, \quad (9)$$

where $\gamma \geq 0$ is the quantum mixer strength.

4.4 Local Gibbs Density Matrix

At inverse temperature $\beta > 0$, define the local density matrix

$$\rho_{r,t} = \frac{\exp(-\beta H_{r,t})}{\text{Tr} \exp(-\beta H_{r,t})}. \quad (10)$$

This is a valid density matrix: it is positive semidefinite and has trace one.

4.5 Eigendecomposition Form

Because $H_{r,t}$ is real symmetric, it admits an eigendecomposition

$$H_{r,t} = V_{r,t} \text{diag}(E_{r,t,1}, \dots, E_{r,t,N}) V_{r,t}^\top. \quad (11)$$

The Gibbs eigenvalue weights are

$$w_{r,t,k} = \frac{\exp(-\beta E_{r,t,k})}{\sum_{\ell=1}^N \exp(-\beta E_{r,t,\ell})}. \quad (12)$$

The diagonal probability used by TSP is

$$P_{r,t,c} = (\rho_{r,t})_{cc} = \sum_{k=1}^N w_{r,t,k} (V_{r,t,c,k})^2. \quad (13)$$

Proposition 3. *For every replica r and slot t , the vector $(P_{r,t,1}, \dots, P_{r,t,N})$ is a probability distribution.*

Proof. Since $\rho_{r,t}$ is a density matrix, it is positive semidefinite and has trace one. Therefore each diagonal entry is nonnegative and

$$\sum_{c=1}^N P_{r,t,c} = \sum_{c=1}^N (\rho_{r,t})_{cc} = \text{Tr}(\rho_{r,t}) = 1.$$

Thus the diagonal entries define a categorical distribution over the city states. $\square$

5 Quantum Entropy and Mixer Energy

5.1 Quantum Entropy

The quantum entropy per replica is

$$S_Q^{(r)} = \sum_{t=1}^N [-\text{Tr}(\rho_{r,t} \log \rho_{r,t})]. \quad (14)$$

Using the eigenvalue weights, this becomes

$$S_Q^{(r)} = - \sum_{t=1}^N \sum_{k=1}^N w_{r,t,k} \log w_{r,t,k}. \quad (15)$$

5.2 Mixer Expectation and Mixer Energy

The mixer expectation is

$$\mathcal{M}^{(r)} = \sum_{t=1}^N \text{Tr}(\rho_{r,t} M). \quad (16)$$

Using the eigendecomposition, this is computed as

$$\mathcal{M}^{(r)} = \sum_{t=1}^N \sum_{k=1}^N w_{r,t,k} v_{r,t,k}^\top M v_{r,t,k}, \quad (17)$$

where $v_{r,t,k}$ is eigenvector k of $H_{r,t}$.

The transverse mixer energy is

$$U_{\text{mixer}}^{(r)} = -\gamma \mathcal{M}^{(r)}. \quad (18)$$

6 Corrected TSP-QEFEM Free Energy

The corrected q-state quantum free energy minimized for each replica is

$$F_Q^{(r)} = U_{\text{tour}}^{(r)} + U_{\text{city}}^{(r)} + U_{\text{mixer}}^{(r)} - \frac{1}{\beta} S_Q^{(r)}. \quad (19)$$

The training loss is the average over replicas:

$$\mathcal{L} = \frac{1}{R} \sum_{r=1}^R F_Q^{(r)}. \quad (20)$$

Proposition 4 (Classical limit at the end of annealing). *When $\gamma = 0$, the mixer energy vanishes and the local Hamiltonian becomes diagonal. The final objective reduces to a classical free-energy form over the probabilities induced by the diagonal Gibbs distributions.*

Proof. If $\gamma = 0$, then

$$H_{r,t} = \text{diag}(a_{r,t}).$$

Therefore $\rho_{r,t}$ is diagonal in the city basis and $U_{\text{mixer}}^{(r)} = 0$. The remaining terms are

$$F_Q^{(r)} = U_{\text{tour}}^{(r)} + U_{\text{city}}^{(r)} - S_Q^{(r)} / \beta,$$

which is the same structural form as a classical mean-field free energy, but reached through a quantum-mixed annealing path. $\square$

7 Hyperparameters Used in the Corrected Berlin52 Run

The corrected run used the following main hyperparameters:

Quantity	Value
Instance	Berlin52
Number of cities N	52
Replicas R	32
Training steps	1500
Data type	float64
Device	CPU
Optimizer	Adam
Learning rate	10^{-2}
City penalty λ_{city}	50.0
$\beta_{\min}$	0.5
$\beta_{\max}$	100.0
$\gamma_{\max}$	0.5
$\gamma_{\min}$	0.0
Mixer length scale τ	0.25
Gradient clipping norm	1.0
Clamp on a	$[-20, 20]$
Known optimum	7542
Vanilla FEM reference best	9410
Corrected QEFEM best	9314

8 Annealing Schedules

The corrected run used exponential beta annealing:

$$\beta_s = \exp \left(\log \beta_{\min} + \frac{s}{N_{\text{steps}} - 1} (\log \beta_{\max} - \log \beta_{\min}) \right), \quad s = 0, 1, \dots, N_{\text{steps}} - 1. \quad (21)$$

The temperature is

$$T_s = \frac{1}{\beta_s}. \quad (22)$$

The mixer strength was linearly annealed:

$$\gamma_s = \gamma_{\max} + \frac{s}{N_{\text{steps}} - 1} (\gamma_{\min} - \gamma_{\max}). \quad (23)$$

Thus the algorithm begins in a high-temperature, mixed regime and ends in a low-temperature, unmixed regime.

9 Chronological Algorithm

The implementation proceeds in the following order.

- Step 1.** Load the TSP coordinates and sort by TSPLIB city label.
- Step 2.** Construct D^{raw} using the TSPLIB EUC_2D integer distance convention.
- Step 3.** Normalize the distance matrix by setting $D = D^{\text{raw}} / \max D^{\text{raw}}$.
- Step 4.** Initialize trainable local diagonal energies $a_{r,t,c}$ near zero.
- Step 5.** Build the distance-aware symmetric mixer M and normalize it by spectral radius.
- Step 6.** For each training step s , set $\beta = \beta_s$ and $\gamma = \gamma_s$.
- Step 7.** Build the local Hamiltonians $H_{r,t} = \text{diag}(a_{r,t}) - \gamma M$.
- Step 8.** Compute the local Gibbs states $\rho_{r,t}$ using eigendecomposition.
- Step 9.** Extract $P_{r,t,c} = (\rho_{r,t})_{cc}$.
- Step 10.** Compute U_{tour} , U_{city} , U_{mixer} , and S_Q .
- Step 11.** Minimize the replica-averaged free energy using Adam.
- Step 12.** Clip gradients and clamp the trainable values $a_{r,t,c}$ to maintain numerical stability.
- Step 13.** Periodically decode each replica using the Hungarian algorithm and record the best raw tour length.
- Step 14.** At the final step, decode all replicas and select the lowest raw tour length.

10 Decoding Rule

At inference time, each replica has a soft assignment matrix

$$P_r \in \mathbb{R}^{N \times N}, \quad P_r[t, c] = P_{r,t,c}.$$

The decoded tour is obtained by solving the linear assignment problem

$$\max_{X \in \{0,1\}^{N \times N}} \sum_{t=1}^N \sum_{c=1}^N X_{t,c} P_{r,t,c} \quad (24)$$

subject to

$$\sum_{c=1}^N X_{t,c} = 1 \quad \text{for all } t, \quad \sum_{t=1}^N X_{t,c} = 1 \quad \text{for all } c. \quad (25)$$

This is implemented with the Hungarian algorithm by minimizing $-P_r$. The output is converted to TSPLIB one-based city labels. The final score is always computed with D^{raw} , not with the normalized matrix D .

11 Result of the Corrected Run

The corrected run produced the following final comparison:

Method	Raw tour length	Gap to 7542	Difference vs. vanilla
Known optimum	7542	0.00%	-1868
Vanilla FEM reference	9410	24.77%	0
Corrected TSP-QEFEM	9314	23.50%	-96

Thus, on this single corrected Berlin52 run,

$$9314 < 9410.$$

The corrected TSP-QEFEM run improved the found raw tour length by 96 units relative to the vanilla FEM reference run. This should be interpreted as a proof-of-life result rather than a statistically final benchmark claim. A reliable research claim requires repeated seeds, matched compute budgets, and ablation studies.

12 Key Takeaways

1. TSP-QEFEM is naturally q-state, not qubit Bloch-state, because each slot chooses among N city labels.
2. The local quantum state is a density matrix $\rho_{r,t} \in \mathbb{R}^{N \times N}$.
3. The diagonal of $\rho_{r,t}$ gives the categorical probabilities used by the TSP objective.
4. The corrected free energy must include the mixer energy U_{mixer} .
5. A distance-aware symmetric mixer is numerically safer than a uniform complete-graph mixer for eigendecomposition-based training.
6. The final discrete tour is obtained by Hungarian decoding and evaluated only with raw TSPLIB distances.